

Auteur: Seda Jusjajeva
 Studierichting: Informatica
 Oefeningenreeks 3G nr. 9.4

$$f(x) = \frac{3x - 8}{4 - 6x}$$

$$\text{dom } f = \mathbb{R} \setminus \left\{ \frac{2}{3} \right\}$$

$$\text{Nulpunten: } 3x - 8 = 0 \Leftrightarrow x = \frac{8}{3}$$

$$\text{Polen: } 4 - 6x = 0 \Leftrightarrow x = \frac{2}{3}$$

VA:

$$\lim_{x \rightarrow \frac{2}{3}^+} \frac{3x - 8}{4 - 6x} = \frac{-6}{0^-} = +\infty$$

$$\lim_{x \rightarrow \frac{2}{3}^-} \frac{3x - 8}{4 - 6x} = \frac{-6}{0^+} = -\infty$$

$$\Rightarrow x = \frac{2}{3}$$

HA:

$$\lim_{x \rightarrow \pm\infty} \frac{3x - 8}{4 - 6x} = \frac{\pm\infty}{\mp\infty}$$

$$\stackrel{H}{=} \lim_{x \rightarrow \pm\infty} \frac{3}{-6} = -\frac{1}{2}$$

$$\Rightarrow y = -\frac{1}{2}$$

x	$-\infty$	$\frac{2}{3}$	$\frac{8}{3}$	$+\infty$
$f(x)$	$-$	$ $	$+$	0

De functie ligt naar $-\infty$ onder de HA en naar $+\infty$ boven de HA.

SA:

geen SA.

Tekenonderzoek:

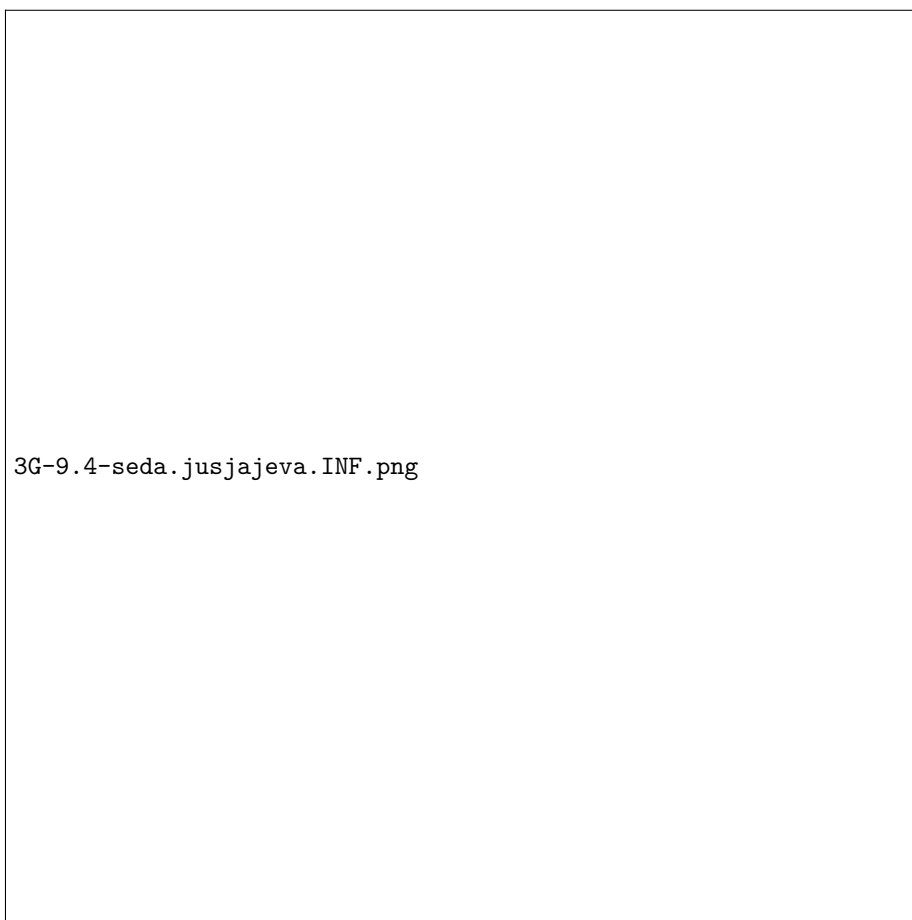
$$\begin{aligned} f'(x) &= \frac{d}{dx} \left(\frac{3x - 8}{4 - 6x} \right) \\ &= \frac{\frac{d}{dx}(3x - 8)(4 - 6x) - (3x - 8)\frac{d}{dx}(4 - 6x)}{(4 - 6x)^2} \\ &= \frac{12 - 18x + 18x - 48}{(4 - 6x)^2} \end{aligned}$$

$$\begin{aligned}
&= -\frac{36}{(4-6x)^2} \\
f''(x) &= \frac{d}{dx} \left(-\frac{36}{(4-6x)^2} \right) \\
&= -36 \cdot \frac{d}{dx} ((4-6x)^{-2}) \\
&= -36 \cdot (-2)(4-6x)^{-3} \cdot \frac{d}{dx}(4-6x) \\
&= \frac{72}{(4-6x)^3} \cdot (-6) \\
&= -\frac{432}{(4-6x)^3}
\end{aligned}$$

x	$-\infty$	$\frac{2}{3}$	$\frac{8}{3}$	$+\infty$
$f(x)$	+	0	-	+
$f'(x)$	-	(2)	-	-
$f''(x)$	-	(3)	+	+
$f(x)$	$\searrow$ (	$\searrow$)	$\searrow$)	$\searrow$)

Grafiek: zie volgende pagina.

References



3G-9.4-seda.jusjajeva.INF.png